



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

SEMESTER 2 2025/2026

SECP3843 – SPECIAL TOPIC IN DATA ENGINEERING

SECTION 1

TUTORIAL ARTICLE

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1.0 The main problem

The main problem of this case study is the lack of a suitable application and/or system to assess and monitor the performance properly, which leads to inefficient decision-making for the top management. This issue has made the company less competitive than its competitors.

2.0 The Main Solution

The main solution for this case study is the development of a Business Intelligence (BI) dashboard to support the PT.MPM to monitor and effectively assess the performance of its business operations. The dashboard was developed by using the Waterfall methodology and deployed using Google Data Studio. The Sales Performance Dashboard and Production Performance Dashboard are its two primary parts. The dashboards display the sales result, revenue trends, forecast accuracy, production output, and the Overall Equipment Effectiveness (OEE).

3.0 User Story

The PT.MPM Business Intelligence Dashboard System will be used by 2 types of users. Table 1 shows the user stories for each user, including the requirements and needs of the users.

Table 1: User Story for PT. MPM Business Intelligence Dashboard System

ID	User Story
US01	As Top management, I want to log in to the dashboard system, so that I can securely access business intelligence dashboard information.
US02	As Top management, I want to view dashboards, so that I can assess and monitor company performance and KPIs.
US03	As Top management, I want analysis dashboard features, so that I can view business information in detail and make informed strategic decisions.
US04	As Top management, I want to share or export reports, so that I can distribute business insights to stakeholders.
US05	As a Data analyst, I want to log in to the dashboard system,

	so that I can manage and analyze company data.
US06	As a Data analyst, I want to upload and update data from the ERP system, so that the dashboard information displayed is accurate and up-to-date.
US07	As a Data analyst, I want to edit dashboard layouts and visualizations, so that dashboards effectively display business information.

4.0 ERD Diagram

Based on the PT.MPM Business Intelligence Dashboard System case study, there are two types of ERD diagrams, which are the Production Data and Sales Data. Figure 1 and Figure 2 illustrate the ERD diagram design using the UML notation.

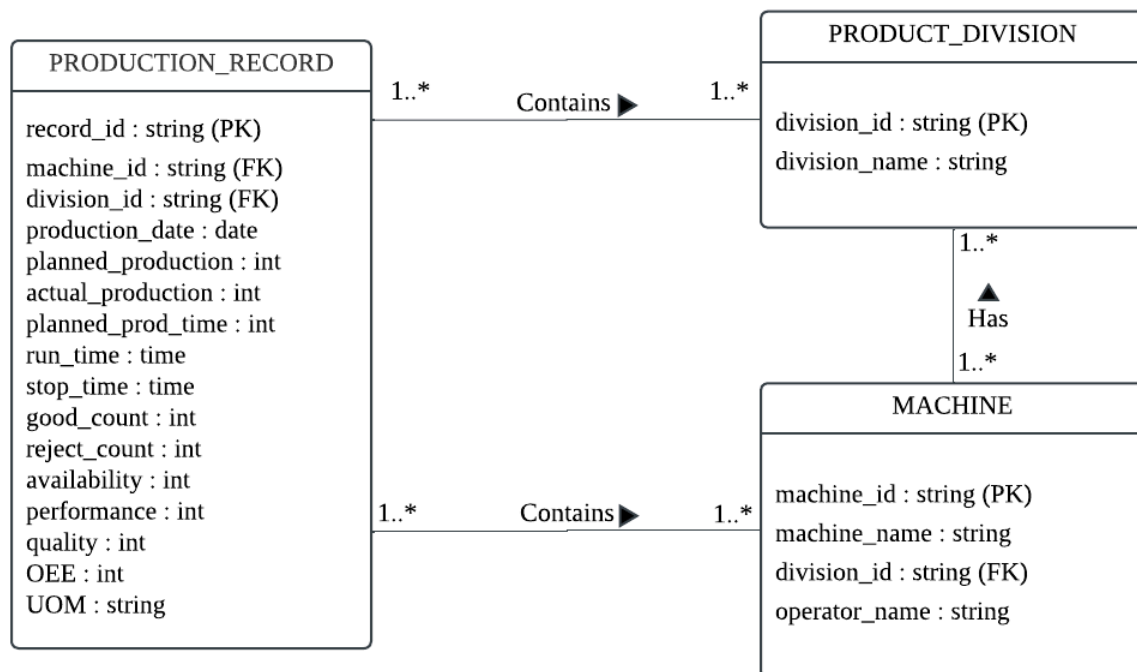


Figure 1: ERD Diagram for Production Data

Figure 1 shows the ERD diagram related to the production data, consisting of relations between three entities. The entities involved are PRODUCTION_RECORD, PRODUCT_DIVISION, and MACHINE.

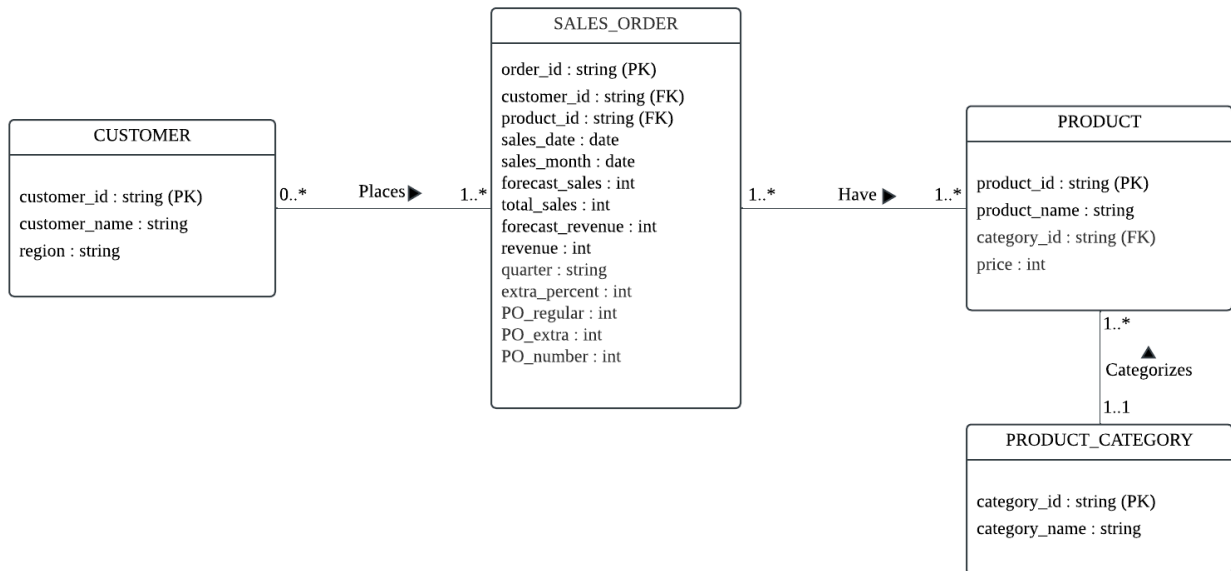


Figure 2: ERD Diagram for Sales Data

Figure 2 shows the ERD Diagram for the sales data, consisting of relations between 4 entities. The entities involved are SALES_ORDER, PRODUCT, PRODUCTION_CATEGORY, and CUSTOMER.

5.0 Use Case Diagram

The case study consists of two users, one external system, and seven use cases. The users are Top Management, who monitor company performance and support decision-making, and Data Analyst, who analyzes business data and design the BI dashboard. The ERP System acts as an external system that supplies data to the BI Dashboard System for reporting and analysis purposes. Figure 3 has defined seven use cases in the case study.

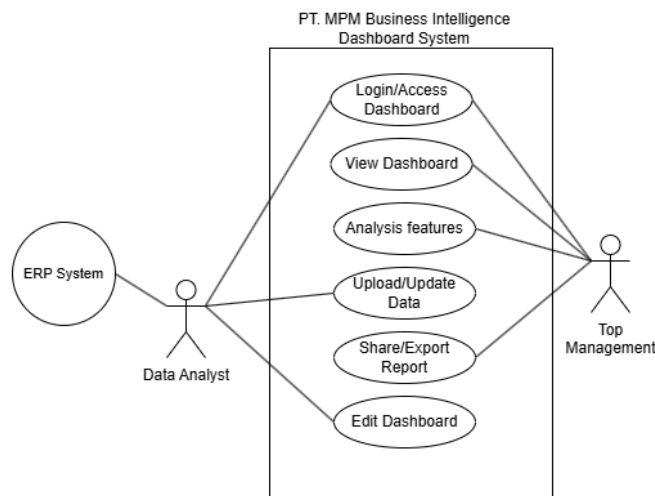


Figure 3: Use Case Diagram for PT. MPM Business Intelligence Dashboard System

6.0 Design Data / System Architecture

Figure 4 illustrates the system and data architecture of the BI dashboard developed for PT. MPM which consists of four main layers.

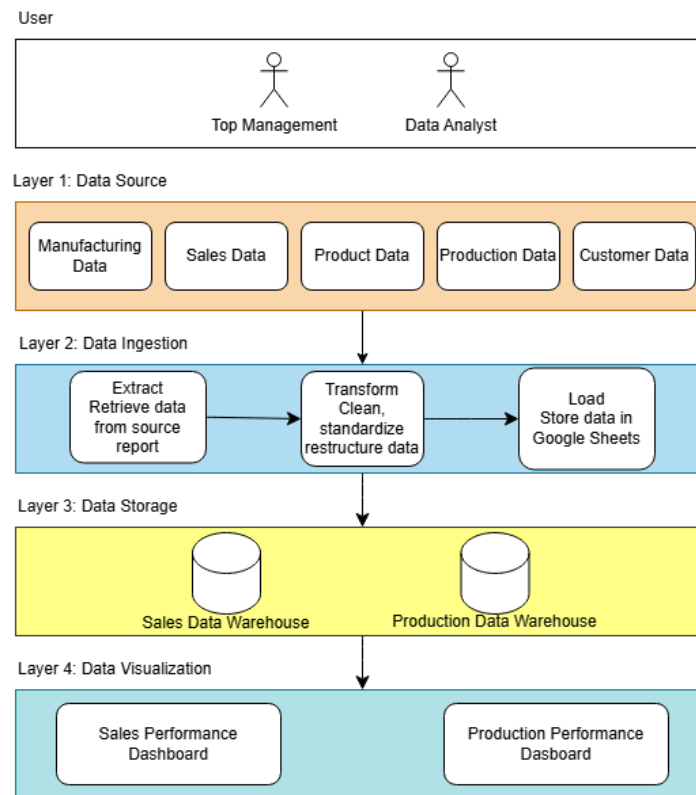


Figure 4: System and Data Architecture of PT. MPM BI Dashboard

Top management and Data Analyst act as the main users who interact with the system to monitor the business performance. The first layer represents the data sources, including manufacturing, sales, product, production, and customer data. These are the raw data collected from different departments within the organization. The second layer is the data ingestion layer where the data is extracted first by retrieving it from source reports, then transformed through cleaning, standardizing and restructuring processes to ensure the data quality and consistency. Finally, the data is loaded into the data warehouse for structured storage.

The data storage layer is divided into two warehouses which are the Sales Data Warehouse and the Production Data Warehouse to support different analytical requirements. Finally, the fourth and the final layer is the presentation layer where the processed data is displayed using two dashboards created in Google Data Studio which are the Sales Performance Dashboards that tracks sales KPIs like revenue and forecast accuracy, and the Production Performance Dashboard that tracks production KPIs like OEE, availability and actual versus planned output. These four layers work together to provide an end-to-end data pipeline that helps PT.MPM's top management efficiently tracks and evaluates the business performance.

7.0 Multidimensional Schema

The multidimensional schema used for the case study is the galaxy schema, which combines 2 fact tables with several dimension tables, as shown in Figure 5.

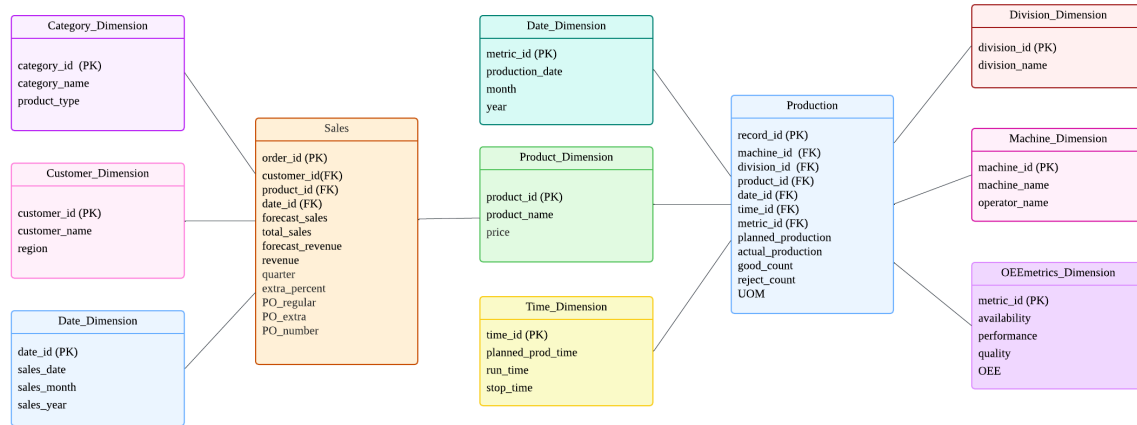


Figure 5: Galaxy Schema of PT. MPM BI Dashboard System

Figure 5 demonstrates the galaxy schema for the case study, consisting of two fact tables for Production and Sales. This galaxy schema involved 9 different dimension tables. Date_Dimension, Time_Dimension, Division_Dimension, Machine_Dimension, OEEmetric_Dimension, Customer_Dimension, Date_Dimension, Product_Dimension, and Category_Dimension are the dimension tables involved in the galaxy schema. The Product_Dimension table acts as a connector that combines the two fact tables.